Free balancing equations worksheet worksheet 12

Maths | 10th grade | Algebra

1. If 3x - 2y = 7 and x + 4y = 11, what is the value of x + y?

A) 3
B) 5
C) 7
D) 9

2. Given the function f(x) = 2x^2 - 4x + 1, what is the value of f(3)?

A) 7
B) 11
C) 13
D) 15

3. If 5^x = 125, what is the value of 2x + 3?

A) 9
B) 11
C) 13
D) 15

4. Solve for x: 2(x - 3) + 3 = x + 5.

A) 4
B) 5
C) 6
D) 7

5. What is the sum of the roots of the quadratic equation 2x^2 - 4x - 6 = 0?

A) 2
B) 3
C) 4
D) 5

6. For which value of x does x^2 - 4x - 5 = 0 have a repeated root?

A) 1
B) 3
C) 5
D) 6

7. If p and q are the roots of x^2 - 3x + 2 = 0, find the value of pq + p + q.

A) 3
B) 5
C) 6
D) 7

8. If y varies directly as x and y = 10 when x = 2, what is y when x = 5?

A) 20
B) 22

C) 25
D) 30

9. If the sequence is defined by a_n = 3n + 2, what is the 7th term?

A) 20
B) 21
C) 22
D) 23

10. If log(base 2) x = 3, what is the value of x?

A) 6
B) 8
C) 10
D) 12

Answer Key

1. B
Solve the system of equations using substitution or elimination to find x = 5 and y = 1, so x + y = 6.

2. C
Substitute x = 3 into the function: f(3) = 2(3)^2 - 4(3) + 1 = 18 - 12 + 1 = 7.

3. D
Since 125 = 5^3, x = 3, so 2x + 3 = 2(3) + 3 = 9.

4. A
Expand and simplify: 2x - 6 + 3 = x + 5. Then x - 3 = 5, hence x = 8.

5. A
The sum of the roots of ax^2 + bx + c = 0 is given by -b/a. Here, -(-4)/2 = 2.

6. D
A quadratic has a repeated root when its discriminant is zero. (-4)^2 - 4(1)(-5) = 16 + 20 = 36, which is not zero.

7. B
pq = c/a = 2/1, p + q = -b/a = 3/1, so pq + p + q = 2 + 3 = 5.

8. A
Direct variation implies y = kx. From y = 10 when x = 2, k = 5. Thus, y = 5(5) = 25.

9. D
Substitute n = 7 into a_n = 3n + 2: a_7 = 3(7) + 2 = 23.

10. B
log(base 2) x = 3 means 2^3 = x, hence x = 8.